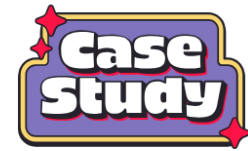


A Case Study on RSA Implementation in Modern Online Banking Systems



ABSTRACT

This study examines asymmetric encryption's role in today's digital safety systems. Take data protection methods - some rely on a pair of distinct keys working together. Information gets locked using one key, known as the public key. Access happens later through another, called the **private key**, held only by authorized users. The process keeps sensitive material secure across transmissions.

A number of existing studies have been conducted to research the many types of asymmetric encryption currently being used in order to secure online banking transactions today. These studies include RSA, ElGamal, and Elliptic Curve Cryptography (ECC), and will be discussed further below.

Introduction

we are all aware of how important it is to protect the information we send, receive, and share on a daily basis through emails, online shopping, and file sharing. Besides that, we know that there exist hackers searching for ways to steal our data. The world of cryptography (which deals with securing information so that only a certain few can have access) mainly offers two types of methods for keeping our data safe: symmetric and asymmetric encryption. **Symmetric encryption, on one hand, involves the use of a single key (or password) for both encrypting and decrypting data; on the other hand, asymmetric encryption (also known as public-key cryptography) requires two different keys, one for encryption and one for decryption**

Asymmetric Encryption



The first known development of cryptographic asymmetrical key algorithms occurred in the mid-to-late 1970s. In contrast to symmetric encryption techniques, which require a private shared secret between parties, asymmetric encryption systems allow for secure communication through the use of one public and one private key. The fact that anyone can use the public key to encrypt messages—while only the

encryption
has two
types

Safee Ullah

holder of the corresponding private key can decrypt those messages—makes asymmetric encryption ideal for use in secure channels such as HTTPS websites and digital signatures.

Literature Review



There are many studies available on asymmetric encryption. One study published in 1978 by Rivest, Shamir, and Adleman presented the RSA algorithm based on the computational difficulty of factoring large prime numbers.

Other literature includes a paper on the ElGamal algorithm introduced in 1985, which uses the discrete logarithm computational difficulty as the basis for the security of encrypted data. Although ElGamal is a valuable encryption method, the ciphertext generated by ElGamal encryption is longer than the corresponding plaintext message.

Various research studies have concluded that asymmetric encryption (such as RSA or ECC) provides better overall security than older asymmetric algorithms, while also being slower compared to symmetric encryptions

Three Types of Asymmetric Encryption

RSA (Rivest-Shamir-Adleman)

One way to lock data uses big math tricks - RSA leads here. Multiply two huge primes, get a number others see but cannot reverse easily. This result becomes part of what people share openly. Locking happens using that shared part. Unlocking needs another piece kept hidden away. Cracking it means splitting the big number again - nearly impossible without those original primes. Hardness hides in how tough division gets at massive sizes..

Basic RSA Steps:

1. Select two prime numbers p and q
2. Compute $n = p \times q$
3. Select e (public exponent)
4. Compute private key d as the inverse of $e \bmod$

id	pass	month
1	no	september
Error nothing happens		
3	yes	april